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Research Article

Chiral Phosphathiahelicenes: Improved Synthetic Approach and Uses in Enantioselective Gold(I)-Catalyzed [2 + 2] Cycloadditions of *N*-Homoallenyl Tryptamines

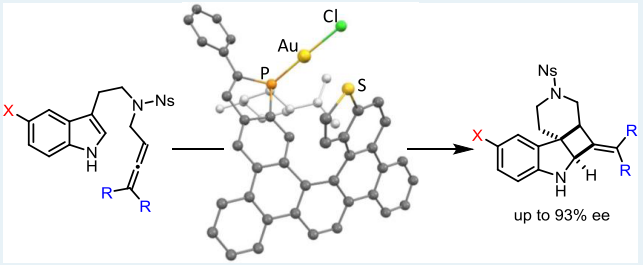
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ABSTRACT: A chiral phosphathiahelicene scaffold displaying a phosphole and a thiophene unit as the terminal rings of the helical sequence has been synthesized and characterized by spectroscopic methods and X-ray diffraction studies. The phosphine oxides (HelPhos-V oxides) have been obtained following a robust and scalable synthetic approach, based on a nickel-promoted alkyne cyclotrimerization reaction. Then, late-stage functionalization has been carried out *via* a bromination/palladium coupling reaction sequence. The HelPhos-V gold(I) complexes have been used as catalysts in the unprecedented enantioselective [2 + 2] cyclization of *N*-homoallenyl tryptamine derivatives, to afford indolenine-fused cyclobutanes in good isolated yields, with enantiomeric excesses up to 93%.



KEYWORDS: *phosphorus ligands, helicenes, gold catalysis, spirofused compounds, tryptamines, indolinecyclobutanes*

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Helicenes and heterohelicenes represent a fascinating class of compounds that have applications in many different fields, including materials, optoelectronics, supramolecular assemblies, biology, and asymmetric catalysis.[1] However, the intrinsic chirality of helical scaffolds has seldom been harnessed for synthesizing phosphorus-containing helicenes for catalytic purposes.[2] This is especially true for phosphahelicenes, *i.e.*, compounds in which phosphorus heterocycles are embedded in the helical scaffold itself. In 2010 and 2012, Tanaka and Nozaki independently reported phosphahelicenes, wherein a phosphole ring was embedded in the internal section of the helical scaffolds (Figure 1a).[3] Since 2012,[4a] our group has developed new families of phosphahelicenes, called "HelPhos-K, -Y, -S, and -P" (Figure 1b), where a five-membered phosphorus ring is embedded at the end of helical structures. This allowed us to access unprecedented chiral tools for organo- and organometallic catalysis. As representative applications, HelPhos-K, HelPhos-S, and HelPhos-P were used notably as ligands in gold(I)-catalyzed allene−ene cyclizations and enyne cycloisomerization processes[4b−d] and HelPhos-Y as a chiral organocatalyst in [3 + 2] cyclizations between 2-arylidenemalononitriles and γ-substituted allenoates.[4e] In most cases, very high levels of catalytic activity and enantioselectivity were reached. Thus, these previous achievements undoubtedly demonstrate the huge potential of HelPhos as chiral ligands and nucleophilic catalysts.

Concerning the strategies for the synthesis of helicene backbones, the most popular has been the oxidative photocyclization of diaryl olefins, developed more than 50 years ago.[5] More recent approaches rely notably on Diels−Alder

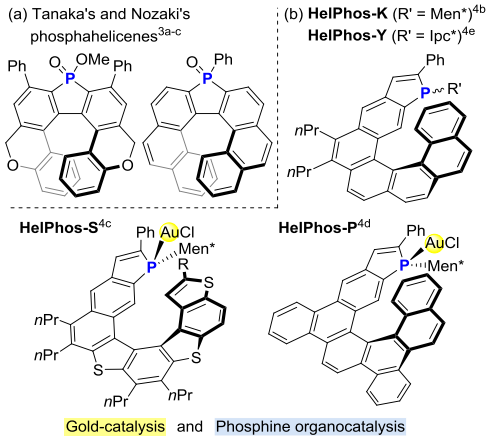


Figure 1. Tanaka's and Nozaki's phosphahelicenes and the HelPhos family (Ipc* = isopinocampheyl; Men* = L-menthyl).

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reactions, metal-catalyzed arylations, cycloisomerizations, and alkyne cyclotrimerizations.[1,6] For the synthesis of phosphahelicenes, we followed two main strategies: the photochemical oxidative cyclization of diaryl olefins[4a−c] and the nickel-catalyzed intramolecular [2 + 2 + 2] cyclotrimerization of triynes.[4d]

So far, the most convenient synthetic approach is the one involving the nickel-catalyzed cyclotrimerization as the key step, which has been applied notably for the synthesis of HelPhos-P.[4d] Compared to the photocyclization-based approach, its main advantages are high yield and ease of scale-up to gram amounts. On the other hand, we have demonstrated, in the HelPhos-S series, that terminal thiophene rings offer a valuable opportunity for late-stage functionalization and easy tuning of the substitution patterns of preformed helical scaffolds. We demonstrated also that the additional substituent on the terminal thiophene ring may have a positive impact on the enantioselectivities in catalytic tests.[7] Therefore, to take advantage of both the cyclotrimerization method and the late functionalization strategy, we have designed a new phosphahelicene scaffold, HelPhos-V, that should be accessible *via* a cyclotrimerization process and displays a terminal thiophene unit suitable for late-stage functionalization (Figure 2).

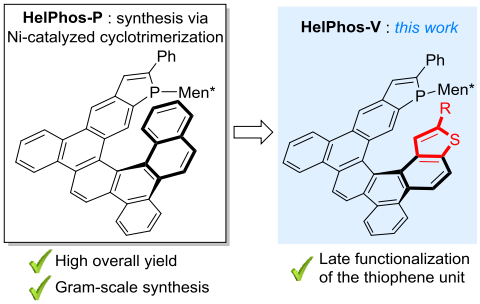
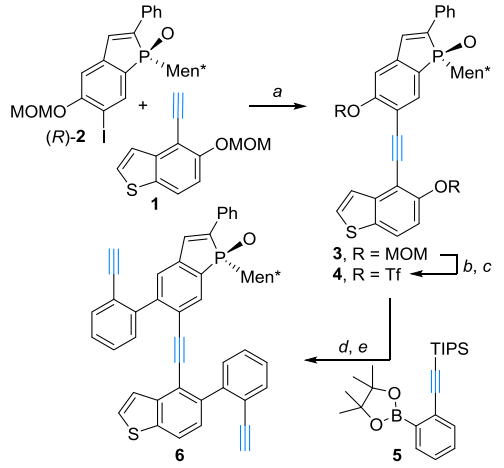


Figure 2. Design of the new phosphathiahelicene HelPhos-V.

In this paper, we will report on the synthesis of HelPhos-V using the cyclotrimerization strategy and the tuning of its substitution pattern. Complexation of the phosphine with Au(I) is reported, as well as the remarkable catalytic activity of the resulting chiral complexes in a new enantioselective gold-catalyzed [2 + 2] cyclization reaction, leading to spiranic indolocyclobutanes.

For the synthesis of the HelPhos-V ligand featuring the cyclotrimerization strategy, we first targeted triyne **6** as the key intermediate (Scheme 1). This compound could be easily obtained in a few steps and on a gram scale *via* a sequence of palladium-catalyzed reactions. First, the methoxymethyl (MOM)-protected 4-ethynylbenzo[*b*]thiophen-5-ol **1** has been prepared in four steps from 5-hydroxybenzothiophene, by regioselective bromination in the C4-position, protection of the alcohol with a MOM group, trans-halogenation (Br → I), and Sonogashira coupling (see the Supporting Information for details). Then, the Sonogashira coupling between **1** and the enantiopure phosphindole (*R*)-**2**[4d] furnished the desired alkyne **3** in a 91% yield (5.2 g). The optimal reaction conditions involved $Pd(PPh_3)_4$/CuI as catalysts, in the presence of diisopropylamine, and heating for 1 h at 80 °C. The next steps include the deprotection of the two MOM groups and the conversion of the phenols into the corresponding triflates for the subsequent metallo-catalyzed

Scheme 1. Synthesis of the Key Intermediate 6



[a]$Pd(PPh_3)_4$ (2 mol %), CuI (4 mol %), diisopropylamine (DIPA), 80 °C, 1 h, 91%. [b]BCl_3, CH_2Cl_2 −78 °C, 0.5 h. [c]$PhNTf_2$, Et_3N, dimethylformamide (DMF), rt, 4 h, 82%. [d]**5**, $Pd(SPhos)_2Cl_2$ (10 mol %), Cs_2CO_3, THF/H_2O (10/1), 80 °C, 1 h. [e]TBAF, THF, rt, 2 h, 79% (two steps).

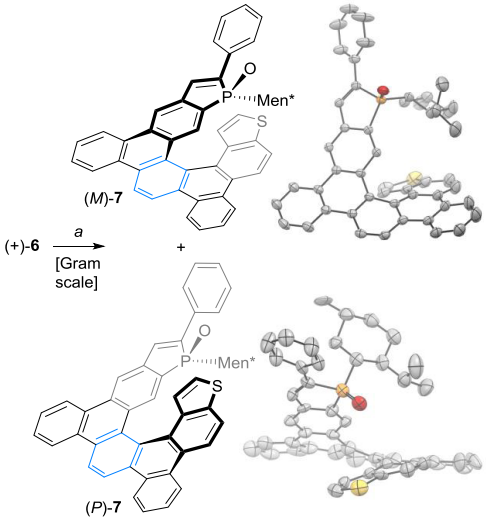
couplings. The seemingly obvious MOM-deprotection step is actually problematic because of the possible addition of the phenol to the alkyne and formation of an undesired benzofurane derivative. After meticulous optimization of this deprotection step with the use of a Lewis acid at a low temperature (BCl_3, CH_2Cl_2, −78 °C) and subsequent triflation with $PhNTf_2$, the desired compound **4** was isolated in an 82% yield over two steps. The following step includes a double Suzuki−Miyaura coupling reaction between the bis-triflate **4** and the boronic ester **5**.[6f] The best reaction conditions consist of a 10 mol % $Pd(SPhos)_2Cl_2$ catalyst, cesium carbonate as a base in a tetrahydrofuran (THF)/H_2O solvent mixture, and heating for 1 h at 80 °C. After deprotection of the triisopropylsilyl ether group (TIPS) with tetrabutylammonium fluoride (TBAF), triyne **6** was finally isolated in a 79% yield (3.4 g) over the last two steps. The next step, *i.e.*, the formation of the phosphahelicene backbone, was accomplished in the presence of $Ni(cod)_2$ (20 mol %) and triphenylphosphine (40 mol %) in THF at room temperature for 30 min (Scheme 2). Starting with 2.4 g of triyne **6**, we isolated 0.91 g of the HelPhos-V oxide (*M*)-**7** {$[\alpha]_D$ − 1099 (c 1, $CHCl_3$), phosphorus-31 nuclear magnetic resonance (^{31}P NMR) δ = 59 ppm} and 1.03 g of (*P*)-**7** {$[\alpha]_D$ + 1635 (c 1, $CHCl_3$), ^{31}P NMR δ = 56 ppm}. These two epimers, with opposite helical configurations, have been obtained in a 47/53 ratio.

With the aim of increasing the (*M*)/(*P*) diastereoisomeric ratio in the cyclotrimerization process, we screened several other monodentate phosphines (PMe_2Ph, PCy_3, P(*p*F-$C_6H_4)_3$), diphosphines (1,1′-ferrocenediyl-bis-(diphenylphosphine) (dppf), 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene (XantPhos)), and triphenylphosphite. Unfortunately, we could not increase the diastereoisomer ratio over a 60/40 ratio. The use of chiral phosphines such as (*R*)-1-(2-diphenylphosphino-1-naphthyl)isoquinoline (QUINAP) and (*S*)-QUINAP,[6f] with a possible beneficial "match effect" between the chiral phosphindole and the chiral catalyst, did not improve these results.

It is quite remarkable that gram quantities of enantiopure phosphathiahelicenes (*M*)-**7** and (*P*)-**7** are available with this

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Scheme 2. Synthesis and X-ray Crystal Structures of (*M*)-7 (CCDC 1984557) and (*P*)-7 (CCDC 1984558)

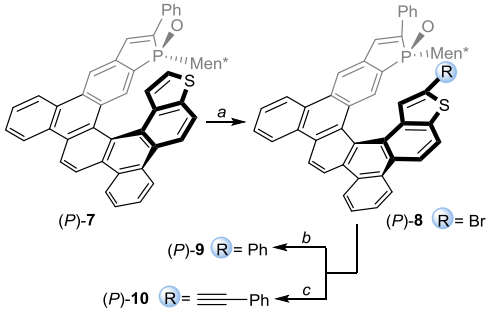


[a]$Ni(cod)_2$ (20 mol %), PPh_3 (40 mol %), THF, rt, 30 min, 81%, (*M*)-7/(*P*)-7 = 47/53.

robust synthetic pathway (48% overall yield starting from compounds **1** and **2**). It is noteworthy that the two epimers have been separated easily by flash chromatography on a silica gel, and the absolute configuration of the helicenes was assigned by X-ray diffraction studies (Scheme 2).

In the next step, we investigated the bromination reaction as a representative example of selective functionalization of phosphathiahelicene (*P*)-7 (Scheme 3). Bromination of the

Scheme 3. Selective Functionalization of Phosphahelicene (*P*)-7



[a]NBS, DMF, rt, 72%. [b]$PhB(OH)_2$, $Pd(PPh_3)_4$ (5 mol %), Na_2CO_3, $THF/H_2O/EtOH$ (2/2/1), 85 °C, 5 h, 81%. [c]PhC≡CH, $Pd(PPh_3)_4$ (8 mol %), CuI (16 mol %), DIPA/DMF (1/1), 80 °C, 16 h, 90%.

terminal thiophene moiety of (*P*)-7 was achieved in a 72% yield with the use of *N*-bromosuccinimide in DMF, leading to α-bromo-thiahelicene **8** as the unique regioisomer. Subsequently, a Suzuki−Miyaura coupling reaction on (*P*)-**8** allowed the introduction of a phenyl group, in an 81% yield (compound **9**). On the other hand, a Pd/Cu-catalyzed Sonogashira coupling furnished the corresponding phenylethynyl-substituted helicene (*P*)-**10** in a 90% yield.

Overall, these experiments confirm that the synthetic strategy benefits from both the high-yielding [2 + 2 + 2] cyclotrimerization approach and the selective and flexible late-stage functionalization of the terminal thiophene ring.

In the past two decades, gold catalysis proved to be a very powerful and versatile tool for the synthesis of densely functionalized compounds. Notably, activation of unsaturated bonds, such as alkynes or allenes, by cationic gold(I) complexes easily delivers structurally complex polycyclic scaffolds.[8] Thus, the demonstrated efficiency of phosphahelicenes as chiral ligands in some of these reactions led us to envisage the conversion of the new HelPhos-V ligands to the corresponding gold(I) complexes and to investigate their catalytic uses.

Phosphine oxide (*M*)-7 was reduced with phenylsilane and the corresponding epimeric mixture of trivalent phosphines[9] was complexed with gold, to isolate the two diastereoisomeric *exo*-(*M*)-**11** {$[\alpha]_D$ − 1444 (c 1, $CHCl_3$), ^{31}P NMR δ = 24 ppm} and *endo*-(*M*)-**11** {$[\alpha]_D$ − 1040 (c 1, $CHCl_3$), ^{31}P NMR δ = 30 ppm} complexes in a 73% yield and a 74/26 ratio (Figure 3a). Both isomers were isolated by column

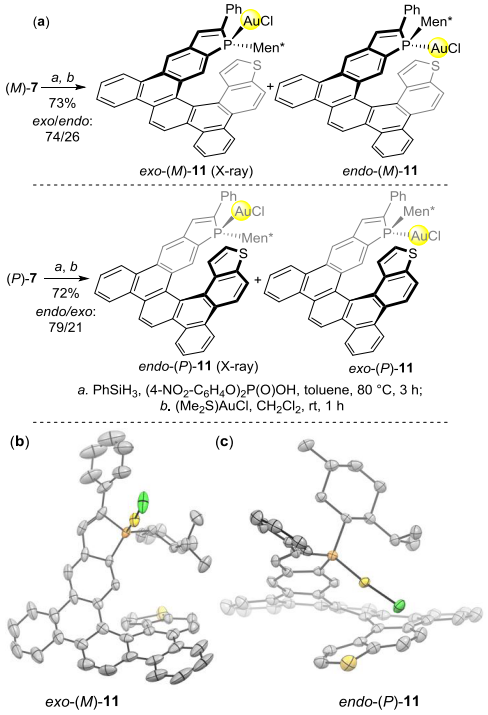


Figure 3. (a) Synthesis of the gold(I) complexes **11**. (b, c) X-ray crystal structures of *exo*-(*M*)-**11** (CCDC 1984559) and *endo*-(*P*)-**11** (CCDC 1984560), respectively. Ellipsoids are drawn at the 50% probability level. H atoms are not shown for clarity.

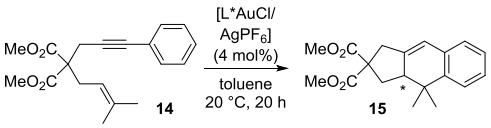
chromatography on a silica gel and fully characterized by NMR (^1H, ^{13}C, ^{31}P). The stereochemistry of the major isomer *exo*-(*M*)-**11** has been assigned by X-ray diffraction studies (Figure 3b). In this complex, the gold atom is located on the external face of the helical scaffold (*exo*-arrangement). Analogously, the reduction−complexation procedure was applied to (*P*)-7, and furnished a 79/21 mixture of phosphahelicene−gold complexes in a 72% yield. X-ray analysis showed that the major isomer displays an *endo*-arrangement (*endo*-(*P*)-**11**), the gold atom being oriented toward the helical pocket (Figure 3c).

Finally, starting from the substituted phosphahelicenes **9** and **10**, the corresponding gold complexes *endo*-(*P*)-**12** and *endo*-(*P*)-**13** were isolated after the phosphine oxide reduction/gold complexation sequence.

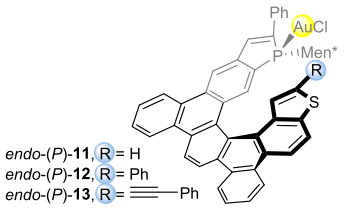
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The newly synthesized chiral gold(I) complexes **11**−**13** were evaluated in a benchmark catalytic reaction, namely, the cycloisomerization of the 1,6-enyne **14** (Table 1).[10] In this

Table 1. Evaluation of the HelPhos-V Ligands in Gold Catalysis



entry	[L*AuCl]	Conv.(%)[a]	ee (%)[b]
1	*exo*-(*M*)-**11**	17	23 (-)
2	*endo*-(*M*)-**11**	39	34 (-)
3	*exo*-(*P*)-**11**	56	37 (+)
4	*endo*-(*P*)-**11**	100	86 (+)
5	*endo*-(*P*)-**12**	45	83 (+)
6	*endo*-(*P*)-**13**	100 (92)	94 (+)



[a]Determined by ^{1}H NMR using trimethoxybenzene as an internal standard. Isolated yield is given in parentheses. [b]Determined by HPLC on a chiral stationary phase.

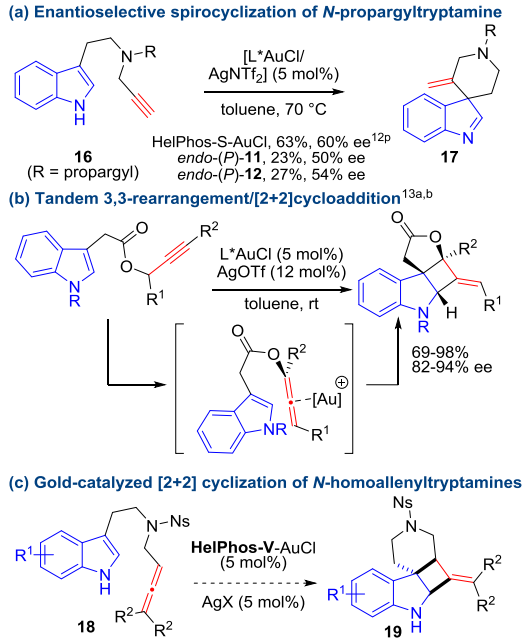
reaction, high enantioselectivity levels were already obtained with bis-gold complexes of axially chiral diphosphines,[10c] mono-gold complexes of chiral phosphites,[10d] and phosphahelicenes.[7]

We first compared the four diastereoisomeric gold complexes *exo*/*endo*-(*M*)/(*P*)-**11** (entries 1−4). In both the (*P*)- and (*M*)-series, the "*endo*" isomers gave better conversions and enantiomeric excesses than the "*exo*" isomers. This can be explained easily by the encapsulation of the metal center in the chiral pocket, in the "*endo*" isomers. The *endo*-cationic gold complex is thus protected by the helical backbone toward possible catalyst decay and it also benefits, in terms of chiral induction, from a highly asymmetric environment. Furthermore, when comparing entries 2 and 4, it appears that a positive synergistic effect exists between the chiral L-menthyl group and the (*P*)-configured phosphahelicene, leading to the best results in terms of enantioselectivity with the *endo*-(*P*)-**11** isomer. In the presence of this precatalyst, and after activation with $AgPF_6$ in toluene at room temperature, the reaction reached full conversion after 20 h and an 86% enantiomeric excess (ee) (entry 4). To optimize this result, with the chiral precatalyst *endo*-(*P*)-**11**, we next screened different silver salts AgX (X = BF_4, OTf, NTf_2, SbF_6) and solvents (xylene, dichloromethane, THF, and acetone). However, neither these modifications nor the decrease of the reaction temperature to 0 °C allowed any improvements (see the Supporting Information for details).

After these preliminary optimization studies, and according to our working hypothesis, we have investigated the possible effects of the remote R-substituents of the thiophene unit on the catalytic behavior. If the introduction of a phenyl substituent in *endo*-(*P*)-**12** does not show positive effects on the enantiomeric excess (83% ee, entry 5), the best result was obtained with the gold(I) catalyst *endo*-(*P*)-**13**, which displays a PhC≡C− substituent on the terminal thiophene ring (92% isolated yield, 94% ee, entry 6). This last result shows undeniably the positive contribution of the additional phenylethynyl substituent on the phosphathiahelicene backbone. This fine-tuning of the ligand allows us to attain the best enantioselectivity obtained so far with this substrate, to the best of our knowledge.[10] Overall, these preliminary catalytic tests demonstrate the potential of the new phosphahelicene scaffold, HelPhos-V, in gold(I) catalysis, both in terms of catalytic activity and chiral induction, and the additional opportunities offered by tuning its substitution pattern.

Next, to take full advantage of this potential, we decided to go beyond model catalytic reactions on known substrates and we turned our attention to some unprecedented gold-catalyzed transformation of indoles. Indeed, indole derivatives are known to afford numerous interesting polycyclic products under gold catalysis.[11,12] Notably, over the last few years, we have been involved in the development of gold-catalyzed spirocyclizations of indoles, leading to a range of spirooxindoles and spiroindolenines.[12o,p] In particular, we disclosed recently the first regioselective spirocyclizations of the *N*-propargyltryptamines **16** into C2-unsubstituted spiroindolenines (Scheme 4a). An enantioselective version of this reaction showed only

Scheme 4. Gold-Catalyzed Cyclizations of *N*-Propargyl and *N*-Homoallenyl-Tryptamine Derivatives



moderate enantioselectivity, with the use of HelPhos-S-AuCl as the gold precatalyst (60% ee).[12p] Unfortunately, the newly developed gold catalysts *endo*-(*P*)-**11** and **12** did not improve this result, highlighting again the challenge offered by this class of reactions when applied to terminal alkynes.

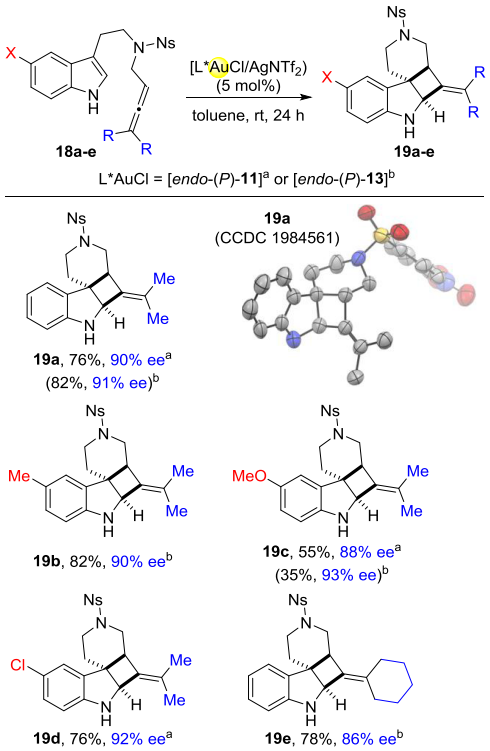
On the other hand, allenes, which display a different coordination mode and geometry, have been more successful in enantioselective gold catalysis involving indoles in hydroarylation reactions. Our attention was in particular driven by the reaction depicted in Scheme 4b that involves an indolyl allene as the key intermediate. It combines a Au(I)-catalyzed 3,3-rearrangement/[2 + 2] cycloaddition of a propargylic ester,

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leading to 2,3-indoline-fused cyclobutanes.[13a] The enantioselective version of this reaction was also developed using chiral acyclic diaminocarbene ligands.[13b,14] We hypothesized that tryptamines containing an *N*-homoallenyl fragment, such as **18**, might undergo analogous intramolecular [2 + 2] cyclizations under gold catalysis and could afford the 2,3-indolinecyclobutanes **19** (Scheme 4c).

We were pleased to see that our first attempts using complex *endo*-(*P*)-**11** and $AgNTf_2$ afforded the desired polycyclic indoline-fused cyclobutane **19a** in a good 76% yield and 90% ee (Scheme 5). A quick screening of the reaction conditions

Scheme 5. [HelPhos-V-AuCl]-Catalyzed Cyclizations of Substrates 18a−e

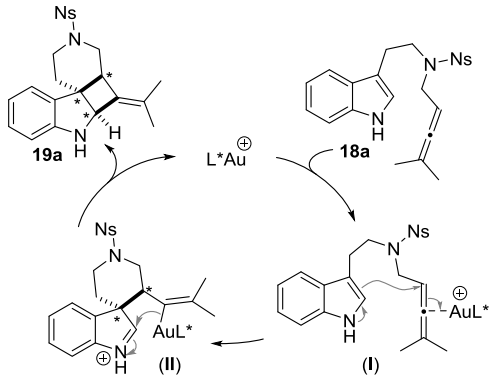


revealed that $AgNTf_2$ was the most suitable chloride scavenger of the gold precatalyst for reactions carried out in toluene at room temperature. The product **19a** was fully characterized by NMR spectroscopy and the skeletal arrangement and the absolute stereochemistry were ascertained by X-ray diffraction studies.

Further experiments have been conducted, then, with the phenylethynyl-substituted gold catalyst (*P*)-**13**, and the corresponding tetracyclic product **19a** was isolated in an 82% yield and 91% ee (Scheme 5). Based on these highly promising results, the scope of the reaction has been extended to a few other substrates, using either precatalyst *endo*-(*P*)-**11** or *endo*-(*P*)-**13**. Modification of the indole moiety with either electron-donating or electron-withdrawing groups gave the corresponding products **19b−d** in moderate to good yields and up to 93% ee.[15] In some instances, the yield could be slightly increased with the use of the *endo*-(*P*)-**11** catalyst (see compound **19c**), but generally, the best compromise between yield and ee was obtained with (*P*)-**13**. Finally, a cyclohexylidene−allene derivative was employed in the Au-catalyzed transformation to furnish the desired product **19e** in a 78% yield and 86% ee.

Concerning the mechanism of this transformation, we can postulate that it starts with the activation of the allene function of substrate **18** by the cationic gold(I) complex (Scheme 6,

Scheme 6. Proposed Mechanism



intermediate **I**). Formation of the first carbon−carbon bond by the nucleophilic addition of indole to the activated allene leads to the spiranic intermediate **II** with concomitant dearomatization of the indole. Analogous 6-*exo* cyclizations involving nucleophilic indoles and tethered unsaturated functions, activated by gold, are quite common with this class of tryptamine derivatives.[11,12] A consecutive intramolecular addition of the vinyl−gold species to the indolenium electrophilic site then affords the 2,3-indoline-fused cyclobutane product **19**, while releasing the active gold complex.

In summary, we have synthesized and characterized new phosphathiahelicenes and used them successfully as chiral ligands in a new gold-catalyzed transformation. Structurally complex polycyclic indole derivatives, with three contiguous stereogenic centers, were isolated in good yields and up to 93% ee. The scalable and versatile synthesis of [HelPhos-V] ligands will enable us to finely tune the phosphahelicene ligands for further applications in enantioselective gold catalysis.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acscatal.0c01819.

General information; experimental procedures; NMR spectra (^1H, ^{13}C, and ^{31}P NMR); HPLC data; and X-ray crystal structure determination (PDF)

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Author Contributions

All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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■ DEDICATION

This article is dedicated to Dr. Jean-René Hamon (Rennes Institute of Chemical Sciences, France) on the occasion of his 65th anniversary.

■ ABBREVIATIONS

MOM, methoxymethyl; $PhNTf_2$, phenyltriflimide; SPhos, 2-dicyclohexylphosphino-2′,6′-dimethoxybiphenyl; TIPS, triisopropylsilyl ether group; TBAF, tetrabutylammonium fluoride; cod, 1,5-cyclooctadiene; dppf, 1,1′-ferrocenediyl-bis(diphenylphosphine); XantPhos, 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene; NMR, nuclear magnetic resonance; DMF, dimethylformamide; THF, tetrahydrofuran; QUINAP, 1-(2-diphenylphosphino-1-naphthyl)isoquinoline

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(15) Deprotection of the nosyl group in compound **19b** afforded the corresponding secondary amine **20b** in 57% yield. See Supporting Information for more details.

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